

Publication List

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Journal Articles

2026

- [1] **Zong, C.**; Banihashemi, F.; Petzold, F.; Lang, W. **Graph-based Database Structure and Calculation Framework for Automated Dynamic Life Cycle Assessment of Buildings.** *Automation in Construction* (accepted for publication), 2026.
- [2] Napps, D.; Staudt, J.; Saluz, U.; Chen, X.; Steiner, D.; **Zong, C.**; Deghim, F.; Lang, W.; Geyer, P.; Schnellenbach-Held, M.; Petzold, F.; Borrmann, A.; König, M. **Evaluation of Building Design Variants in Early Phases on the Basis of Adaptive Detailing Strategies.** *Buildings*, 16(4), 685, 2026. doi: 10.3390/buildings16040685.

2025

- [3] **Zong, C.**; Banihashemi, F.; Lang, W. **Dynamic Life Cycle Impact Assessment (DLCIA) in a Sustainable Building Planning Process.** *Scientific Reports*, 15, 32680, 2025. doi: 10.1038/s41598-025-20599-1.

2024

- [4] **Zong, C.**; Chen, X.; Deghim, F.; Staudt, J.; Geyer, P.; Lang, W. **A Holistic Two-Stage Decision-Making Methodology for Passive and Active Building Design Strategies under Uncertainty.** *Building and Environment*, 251, 111211, 2024. doi: 10.1016/j.buildenv.2024.111211.
- [5] Banihashemi, F.; Weber, M.; Deghim, F.; **Zong, C.**; Lang, W. **Occupancy Modeling on Non-Intrusive Indoor Environmental Data through Machine Learning.** *Building and Environment*, 254, 111382, 2024. doi: 10.1016/j.buildenv.2024.111382.

2023

- [6] Sun, Y.; **Zong, C.**; Pancheri, F.; Chen, T.; Lueth, T. C. **Design of Topology Optimized Compliant Legs for Bio-Inspired Quadruped Robots.** *Scientific Reports*, 13, 4875, 2023. doi: 10.1038/s41598-023-32106-5.

2022

- [7] **Zong, C.**; Margesin, M.; Staudt, J.; Deghim, F.; Lang, W. **Decision-Making under Uncertainty in the Early Phase of Building Façade Design Based on Multi-Objective Stochastic Optimization.** *Building and Environment*, 2022. doi: 10.1016/j.buildenv.2022.109729.

Conference Papers

2026

- [8] Quiroz, JV.; Ehlers, N.; **Zong, C.**; Lang, W. **Life Cycle Assessment of urban spaces: a case study on a district scale.** *IOP Conference Series: Earth and Environmental Science*, SBE 2025, Trondheim, 2025. doi: 10.1088/1755-1315/1615/1/012076. (Master's thesis supervision)

2025

- [9] Chen, H.; Benfer, R.; **Zong, C.**; Reitberger, R.; Präger, L.; Lang, W. **A RAG-Based Framework for Material Matching in LCA: Integrating Semantic and Character Similarity with AI-Driven Explanations.** *Building Simulation Conference 2025*, Proceedings of the 19th IBPSA Conference, 2025. doi: 10.26868/25222708.2025.1449.
- [10] **Zong, C.**; Lang, W. **Dynamic Factors in Life Cycle Assessment: Systematic Review and Categorization.** *Building Simulation Conference 2025*, 2025. doi: 10.26868/25222708.2025.1450.
- [11] Ehlers, N.; **Zong, C.**; Meier-Dotzler, C.; Lang, W. **Comparing Refurbishment Strategies for Buildings at a City District Level Using Life Cycle Analysis.** *Journal of Physics: Conference Series*, CISBAT 2025, Lausanne, 2025. doi: 10.1088/1742-6596/3140/13/132010.
- [12] Kobrouslı, M.; **Zong, C.**; Ehlers, N.; Lang, W. **Concept Paper for a Python Tool to Estimate Recycle and Reuse Potential of Building Materials.** *IOP Conference Series: Earth and Environmental Science*, SBE 2025, Zurich, 2025. doi: 10.1088/1755-1315/1554/1/012003. (Master's thesis supervision)
- [13] **Zong, C.**; Sun, Y. **Assessing Pedestrian Level Barriers Using Mobile Robot to Validate 15-Minute City Concept.** *Building Simulation Conference 2025*, 2025. doi: 10.26868/25222708.2025.1448.

2024

- [14] Beltinger, A.; Ehlers, N.; **Zong, C.**; Lang, W. **Gebäudetypen für Nicht-Wohngebäude zur vereinfachten Durchführung von Lebenszyklusanalysen auf Stadtquartiersebene.** *BauSim Conference 2024*, 10th Conference of IBPSA-Germany and Austria, 2024. doi: 10.26868/29761662.2024.39. (Master's thesis supervision)
- [15] **Zong, C.**; Sun, Y.; Lang, W. **System Dynamics Modeling of Life Cycle Carbon Footprints for Building Wall Insulation Materials.** *IOP Conference Series: Earth and Environmental Science*, 1363, WSBE 2024. doi: 10.1088/1755-1315/1363/1/012066.

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- [16] Ehlers, N.; Schulze, K.; **Zong, C.**; Vollmer, M.; Schroeter, S.; Lang, W. **A Holistic Analysis of Sustainability Metrics at an Urban District Scale.** *IOP Conference Series: Earth and Environmental Science*, SBE23, Thessaloniki, 2023. doi: 10.1088/1755-1315/1196/1/012071.

- [17] **Zong, C.**; Reitberger, R.; Deghim, F.; Staudt, J.; Larkin, J.A.; Lang, W. **A Comparative Study of Dynamic Building Simulation and Machine-Learning within a Two-Stage Multi-Objective Stochastic Optimization Framework.** *18th IBPSA Conference on Building Simulation*, BS 2023. doi: 10.26868/25222708.2023.1369.

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- [18] Staudt, J.; Margesin, M.; **Zong, C.**; Deghim, F.; Lang, W.; Zahedi, A.; Petzold, F.; Schneider-Marín, P. **Life Cycle Potentials and Improvement Opportunities as Guidance for Early-Stage Design Decisions.** *ECPPM 2022*, 14th European Conference on Product and Process Modelling, 2022. doi: 10.1201/9781003354222-5.
- [19] Steiner, D.; Schnellenbach-Held, M.; Staudt, J.; Margesin, M.; **Zong, C.**; Lang, W. **Structure and LCA-Driven Building Design Support in Early Phases Using Knowledge-Based Methods and Domain Knowledge.** *ECPPM 2022*, 14th European Conference on Product and Process Modelling, 2022. doi: 10.1201/9781003354222-6.
- [20] **Zong, C.**; Banihashemi, F.; Vollmer, M.; Lang, W. **Implementation of Occupant Behaviour Models for Window Control Using Co-Simulation Approach.** *BauSIM 2022*, 2022. doi: 10.26868/29761662.2022.85.